

SHIELD USA: Leap-ahead Organic Substrate Enabled by Fan-Out Technology, Domestic Materials, Process Innovations, and Novel EDA Workflows

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Abstract — This paper will present the objectives and technical progress of the SHIELD USA (Substrate-based Heterogeneous Integration Enabling Leadership Demonstration for the USA) program funded under the CHIPS and Science Act. The SHIELD USA vision is to design and develop molded core substrates that enable leap-ahead performance in organic substrate microelectronic packaging. SHIELD USA will enable a domestic ecosystem of new materials, processes, equipment and designs proven on a series of demonstrators and product exemplars. Key challenges and status of all developments across design efforts, EDA software tool development, and process engineering, are highlighted in the paper.

Keywords — SHIELD USA, organic substrate, molded fan-out core, VIB, planarized wiring, EDA, heterogeneous integration

I. INTRODUCTION

This paper will present the goals, and for the first time, technical progress of the SHIELD USA (Substrate-based Heterogeneous Integration Enabling Leadership Demonstration for the USA) program funded under the CHIPS and Science Act. The vision to design and develop molded core substrates that enable leap-ahead performance in organic substrate microelectronic packaging as shown in Fig. 1 will be laid out. SHIELD USA molded-core substrates are manufactured using Deca’s fan-out technology [1] with embedded active and passive devices (e.g., inductors and capacitors), fine pitch through substrate core vias, and dual-sided redistribution layer (RDL) wiring on build-up dielectric films (BUF). SHIELD USA will enable a domestic ecosystem of new materials, processes, equipment and designs proven on a series of demonstrators and product exemplars.

II. SHIELD USA ROADMAP

SHIELD USA will develop process demonstrators to show performance against technical targets as outlined in Fig. 2. We summarize the status and challenges of the engineering developments against Phase 1 targets. SHIELD USA, under the leadership of ASU and Deca, has established a strong team of partners to deliver crucial technologies to the National Advanced Packaging Manufacturing Program (NAPMP). In the next sections, we will mainly focus on the process engineering developments by ASU and Deca and on the electronic design automation (EDA) software developments led by Synopsys.

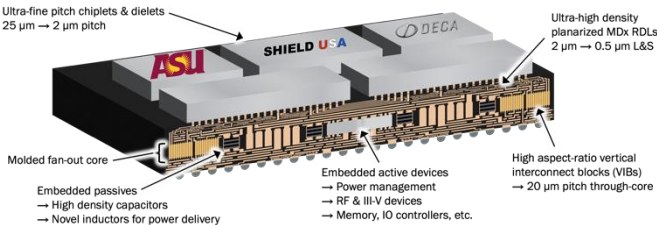


Figure 1. Schematic cross-section of SHIELD USA’s molded fan-out core substrate with embedded passive and active devices, fine pitch high aspect ratio through-core vias, and dual-sided wiring based on fan-out technology

Demonstrator	1	2	3	4
Min. Line Width	2 μm	1.5 μm	1 μm	0.5 μm
Min. Line Space	2 μm	1.5 μm	1 μm	0.5 μm
Min Bump Pitch	25 μm	15 μm	8 μm	2 μm
Interlevel Via Diameter	5 μm	3 μm	2 μm	1 μm
Thru-Substrate Via Diameter	35 μm	25 μm	20 μm	12 μm
Max # Layers Per Side	3	5	7	10
Max Thickness	0.85 mm	1.25 mm	1.5 mm	1.5 mm
Embedded Passives		Capacitors	Capacitors + Inductors	Capacitors + Inductors

Figure 2. SHIELD USA roadmap across four demonstrators and phases

III. SHIELD USA PROCESS ENGINEERING PROGRESS

SHIELD USA introduces several key process features that enable unique design opportunities for AI and high-performance computing (HPC). The next paragraphs summarize progress for a) vertical interconnect blocks (VIBs) b) molded fan-out core and c) ultra-high density planarized wiring layers.

VIBs enable a vertical interconnect which is not AR limited, as it is the case for traditional approaches of through-mold vias, mechanical or laser drilled vias and through-silicon or through-glass vias. VIBs offer high aspect ratios ($AR > 15$) vertical through connections. Fabricated VIBs are shown in Figure 3. The VIB multi-layer wiring is first fabricated on a planar wafer (Fig. 3a), singulated into blocks of interconnects (Fig. 3b), rotated by 90 degrees during placement (Fig. 3c), and then embedded into the molded core substrate to form through-substrate connections (Fig 4). Because the VIB wiring is first built laterally on a wafer, critical dimensions of $\leq 20\mu\text{m}$ with non-rectilinear patterns are feasible.

SHIELD USA’s molded fan-out core serves as a functional core for the substrate which enables the embedding of capacitors, inductors, dies, and VIBs as shown in Fig. 1.

Embedding passive components into the substrate core with low-resistivity and thick Cu connections on both top and bottom surfaces are imperative for vertical power delivery applications. Early prototypes of molded fan-out substrate cores with embedded passives (capacitors) have been fabricated and are highlighted in Fig. 5.

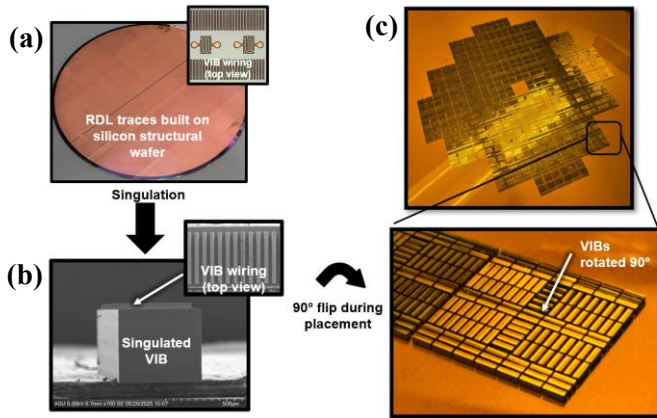


Figure 3. (a) Fabricated, (b) singulated and (c) placed Vertical Interconnect Blocks at wafer scale

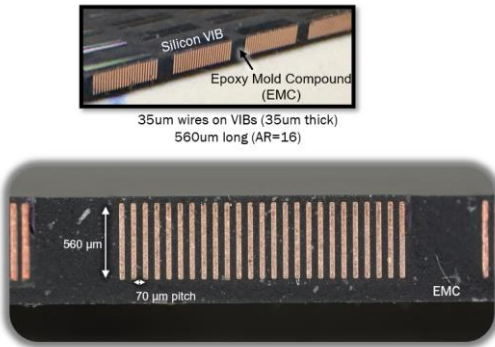


Figure 4. Vertical Interconnect Blocks embedded in Epoxy Mold Compound enabling a vertical through connection of AR=16 through the SHIELD USA substrate core



Figure 5. Embedded capacitive components into the substrate core with low-resistivity, thick Cu connections on both top and bottom surfaces

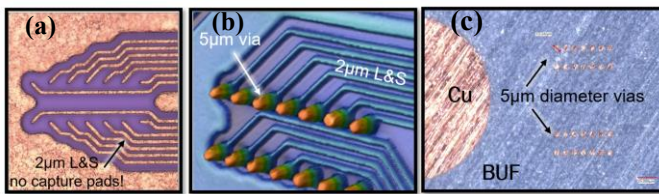


Figure 6. (a) Ultra-high density planarized wiring layers at $2\mu\text{m}$ L&S and (b) $5\mu\text{m}$ diameter vias plated on $2\mu\text{m}$ traces without capture pads. (c) The RDL is then encapsulated by the dielectric BUF, the wafer is planarized and the $5\mu\text{m}$ vias are revealed

Dual-sided redistribution layer (RDL) wiring surrounding the molded core substrate is an important building block for delivering multi-layer SHIELD USA substrates as per Fig. 1. SHIELD USA is also developing the first domestically fabricated build-up film (BUF) to be used as the insulative dielectric for the substrate layers. Ultra-high density planarized wiring layers employing Deca's MDx™ process have been demonstrated and encapsulated by the develop BUF at $2\mu\text{m}$ lines & spaces (L&S) (see Fig. 6a). Along with the redistribution wiring, scaled $5\mu\text{m}$ diameter vias have been demonstrated (see Fig. 6b). Unlike traditional photopolymer vias, MDx vias are defined by additively electroplated Cu vias in photoresist *over each wiring layer*. The vias are then encapsulated with the build-up film dielectric and planarized to reveal (see Fig. 6c). Therefore, topography is limited to the planarization tolerance. Since these vias are defined by photoresist with high resolution and excellent sidewalls, capture pads are not required (see Fig. 6a), significantly increasing routing density. The planarized structure also allows via stacking without inherent process limitations.

To ensure manufacturing robustness for such scaled features, we are employing Deca's Adaptive Patterning® (AP) technology [1] and we are developing new software capabilities to extend AP to Adaptive Defect Mitigation (ADM). Deca's ADM technology will mitigate defects *during* build-up by employing automatic optical inspection after each layer fabrication and by dynamically rerouting around classified defects on *subsequent* layers. Currently, ADM is demonstrated on designs with simulated defects. ADM demonstration on real defects and novel design methodologies with supporting electronic design automation (EDA) are part of the future SHIELD USA phases. Note that this approach does not require circuitry on die, minimizing the overhead in current die-to-die interfaces which typically use redundant lanes.

IV. SHIELD USA EDA DEVELOPMENTS

Historically, integrated circuit (IC) design and package design have operated as distinct domains, each managed by separate teams using different toolsets. The traditional IC design flow spanning system specification, architectural and functional design, physical implementation, and signoff has relied on highly automated EDA tools. In contrast, package design has remained semi-automated, focusing primarily on floorplanning, pinout, placement, routing, and verification, with limited integration to the IC design process. As system complexity increases and heterogeneous integration becomes mainstream, this separation is no longer tenable, necessitating a unified approach where IC, interconnect, and package design are tightly integrated from the outset and supported by a comprehensive Process and Assembly Design Kit (PADK) to ensure accuracy, manufacturability, and co-optimization.

To address these challenges, the collaboration between Deca, ASU and Synopsys advance design and verification of next-generation fan-out packages. Synopsys 3DIC Compiler [2] brings EDA automation to Deca's fan-out technology, enabling rapid design exploration and unified physical design flows.

Designers will estimate critical parameters such as layer count, feature dimensions, and package footprint during early architectural definition, facilitating evaluation of alternative package structures. The tool will support simultaneous floorplanning of chiplets and packages, leveraging IC-style design rule-aware auto-routing to streamline interconnect design. Crucially, interconnect modeling, timing closure, and comprehensive verification will be performed in a manner analogous to IC metal layers, utilizing established power, signal integrity, and thermal analysis methodologies. This integrated workflow allows for full system signoff, encompassing both chiplets and package, and ensures robust coverage of assembly and packaging rules.

The extension of 3DIC Compiler to support molded fan-out core substrates will advance multi-die integration. Designers will be able to explore and partition package and dies, advanced die-to-die (D2D) routing for standards such as HBMx and UCIe, and in-depth analysis including parasitic extraction and static timing signoff. The platform as represented in Fig. 7 will enable detailed planning for die stacking, component floorplanning, connectivity, bump/VIB placement, and power delivery, while integrated signal and power integrity analysis ensures system reliability. Ultimately, this holistic approach combining rapid exploration, unified implementation, and analysis-driven signoff will empower the development of highly complex, high-performance systems.

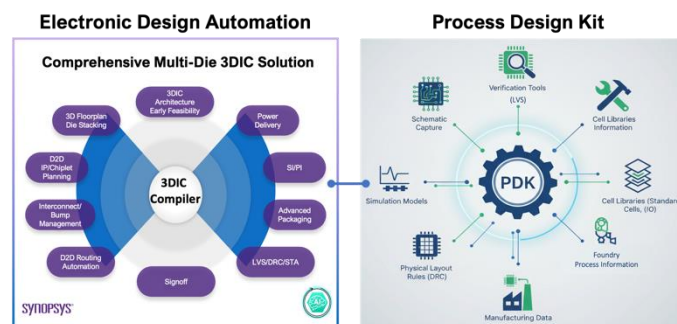


Figure 7. SHIELD USA Integrated Design Automation Platform

SYNOPSIS 3DIC COMPILER

Synopsys 3DIC Compiler is an end-to-end electronic design automation platform that unifies architectural exploration, physical implementation, and system level analysis for advanced multi-die integration in both 2.5D and true 3D configurations. It explicitly targets organic substrate solutions and multi-chiplet topologies—providing configuration, prototyping, and validation capabilities for each. By operating on a common design data model shared across the Synopsys design family, the tool consolidates previously disjoint workflows into a single environment that spans planning, design, verification, and signoff. This integration is aimed at accelerating design closure while preserving accuracy across heterogeneous process nodes and extremely high inter-die connectivity densities. Its “architecture-to-signoff” continuum reduces iteration count and preserves intent across stages.

3DIC Compiler features immersive visualization: designers can fluidly move between 2D and true 3D views of the system, using cross-probing to correlate logical hierarchy with physical geometry across dies and package levels. Interactive bump planning, TSV pad definition, and interposer connectivity inspection are embedded directly in the GUI, supported by native 3D design-rule checking to ensure alignment, spacing, and enclosure constraints are met in multi-die contexts.

For physical implementation, the platform includes high-throughput engines for automated D2D routing on interposers or base dies, incremental ECO, and DFx flows, all governed by native 3D DRC/LVS capabilities that understand multi-die connectivity semantics. These checks operate across die boundaries, validating connectivity and physical integrity for stacked and interposer-based assemblies alike.

3DIC Compiler is designed to converge system performance with integrated analysis across timing, power, reliability, and thermal domains. Tight coupling with Synopsys signoff tools—PrimeTime for static timing analysis, StarRC for parasitic extraction, and IC Validator for physical verification—enables design teams to iterate rapidly while maintaining “golden” accuracy. In addition, the platform integrates Ansys RedHawk-class multiphysics engines to model signal integrity (SI), power integrity (PI), electromigration/IR (EM/IR), and thermal behavior for the *full* multi-die system, not just individual dies. This integrated co-analysis accelerates convergence and reduces the risk of late-stage surprises in stacked or chiplet-based designs.

PADK: PROCESS AND ASSEMBLY DESIGN KIT

In multi-die design, a PDK encodes semiconductor process rules and device/cell models for wafer-level implementation, whereas an ADK encodes assembly/package constraints, manufacturability rules, and sign-off collateral for interposers, fan-out substrates, organic laminates, bridges, and die-to-die interfaces. A PADK is an extension of a PDK that incorporates the ADK. A PADK is consumed by EDA flows to ensure that package-level geometry, connectivity, and materials comply with manufacturing limits. The PADK must be interoperable with 3D design platforms (e.g., 3DIC tools) and verification engines so that assembly-level rules (die spacing, bump pitch/density, keep-outs, TSV alignment, enclosure constraints) can be enforced and analyzed alongside timing, extraction, and multiphysics sign-off.

A production PADK combines technology descriptions, rule decks, libraries, and reference flows that collectively enable “correct-by-construction” package implementation:

- **Technology files for package/interposer** (e.g., *tech file* .tf): layer stacks (RDL metals, vias, pillars), design rules, and physical parameters used by the 3D implementation cockpit.
- **Layer-mapping collateral** (e.g., *TF2GDS* mapping) and **reference libraries** (e.g., Synopsys NDM bump/ball/TSV cells) to standardize IO constructs and export consistency.
- **Physical-verification runsets** (IC Validator **DRC/LVS/FILL** .rs): assembly checks across dies/interposers (die-to-die spacing, alignment, enclosure,

connectivity, density), including inter-die interface legality decks.

- **Parasitic-extraction technology files** (StarRC ITF/NXTGRD + .cmd): resistivity/capacitance models for RDL/stack metals and vias, plus foundry-recommended extraction settings.
- **Assembly rules runsets** (inter-die DRC/LVS): constraints for spacing, pillar-to-pillar clearance, seal ring and keep-out regions, etc., enforced at the package/system level.
- **Standardized system descriptions** (e.g., 3Dblox/MDXF): machine-readable multi-die stack syntax capturing chiplet orientation, stack topology, connections, feasibility constraints, test probes, and alignment markers to drive automated checks and cross-tool exchange.
- **Reference flows and exemplars**: scripted methodologies demonstrating construction, routing, verification, and co-analysis across dies and package (including RF/AI exemplars).

EDA CONTRIBUTION SUMMARY

In contrast to piecemeal flows that stitch together 2D tools for 3D problems, 3DIC Compiler offers a hyper-converged environment expressly built for multi-die systems. The approach reduces the friction between planning, implementation, and signoff, shortens learning curves through a familiar GUI and command set aligned with IC Compiler II/Fusion Compiler, and supports rigorous, system-level integrity analysis needed for production-grade 3D designs. As more organizations adopt chiplet strategies and fine-pitch interposer technologies, such an integrated platform becomes central to achieving robust convergence on SI/PI/thermal constraints while meeting aggressive schedule and cost targets.

PADKs provide the packaging rule infrastructure required to automate and sign off multi-die integration at scale. Their contents—technology files, verification decks, extraction tech, standardized system descriptions, and reference flows—are authored against the packaging DRM and validated via exemplar designs and auto-generated interface checks. Complexity arises from heterogeneous materials, multi-physics coupling, cross-PDK interfaces, and the need for cross-vendor interoperability. As 3DIC platforms increasingly unify exploration-to-signoff, PADKs become the linchpin that enables correct-by-construction assembly while maintaining accuracy and scalability for dense chiplet systems.

The SHIELD USA EDA effort is building a production-quality PADK that makes advanced molded-core packaging rules-driven, interoperable, and sign-off capable. By unifying a single 3D cockpit in Synopsys' 3DIC Compiler—and by standardizing stack descriptions—the program reduces iteration, mitigates risks, and establishes a sustainable ecosystem handoff for education, research, and deployment.

V. CONCLUSION

The SHIELD USA program has demonstrated initial progress toward advanced organic substrate integration through its structured roadmap and collaborative engineering efforts. Key developments include the fabrication of high aspect ratio vertical interconnect blocks, molded fan-out core substrates, and the implementation of ultra-high density planarized wiring, contributing to enhanced electrical performance and integration capabilities. The phased approach has enabled systematic evaluation of materials, process steps, and design methodologies within a domestic ecosystem. Ongoing work will focus on further optimizing these technologies, addressing feature scaling, reliability and manufacturability, and expanding unified EDA workflows to support increasingly complex heterogeneous integration.

ACKNOWLEDGEMENT

This work is supported by the Department of Commerce and NIST under Contract #80NANB25T050. The views and conclusions contained herein are those of the authors and should not be interpreted as necessarily representing the official policies or endorsements, either expressed or implied, of the Department of Commerce and NIST or the U.S. Government. The U.S. Government is authorized to reproduce and distribute reprints for Governmental purposes notwithstanding any copyright notation herein.

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